MOHD AYAN HUSSAIN
ROLL NO. 1250258271
BCADS14

```python
# ================================
# Netflix Data Analysis Assignment
# ================================

# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("netflix.csv")

# Preview data
df.head()
```

```
  show_id     type                   title         director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show          Blood & Water              NaN
2      s3  TV Show              Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show           Kota Factory              NaN

                                                   cast        country  \
0                                                   NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...            NaN
3                                                   NaN            NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...          India

           date_added  release_year rating   duration  \
0  September 25, 2021          2020  PG-13     90 min
1  September 24, 2021          2021  TV-MA  2 Seasons
2  September 24, 2021          2021  TV-MA   1 Season
3  September 24, 2021          2021  TV-MA   1 Season
4  September 24, 2021          2021  TV-MA  2 Seasons

                                          listed_in  \
0                                    Documentaries
1    International TV Shows, TV Dramas, TV Mysteries
```

```
2  Crime TV Shows, International TV Shows, TV Act...
3                              Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV ...

                                           description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...
3  Feuds, flirtations and toilet talk go down amo...
4  In a city of coaching centers known to train I...
```

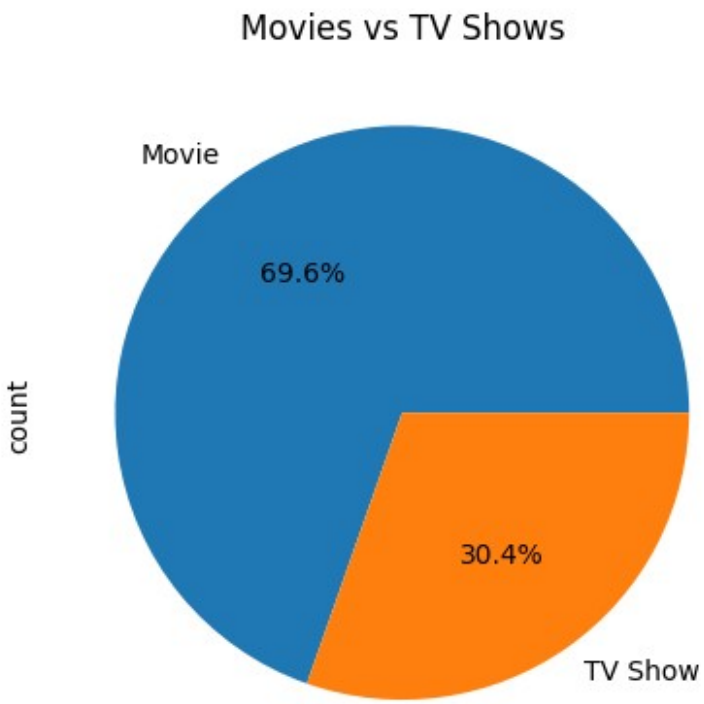
```python
# Basic info
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```python
# Convert date_added to datetime
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')

# Extract year and month
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month

# Q1. Ratio of Movies vs TV Shows
type_counts = df['type'].value_counts()
print(type_counts)

type_counts.plot(kind='pie', autopct='%1.1f%%')
plt.title("Movies vs TV Shows")
plt.show()
```

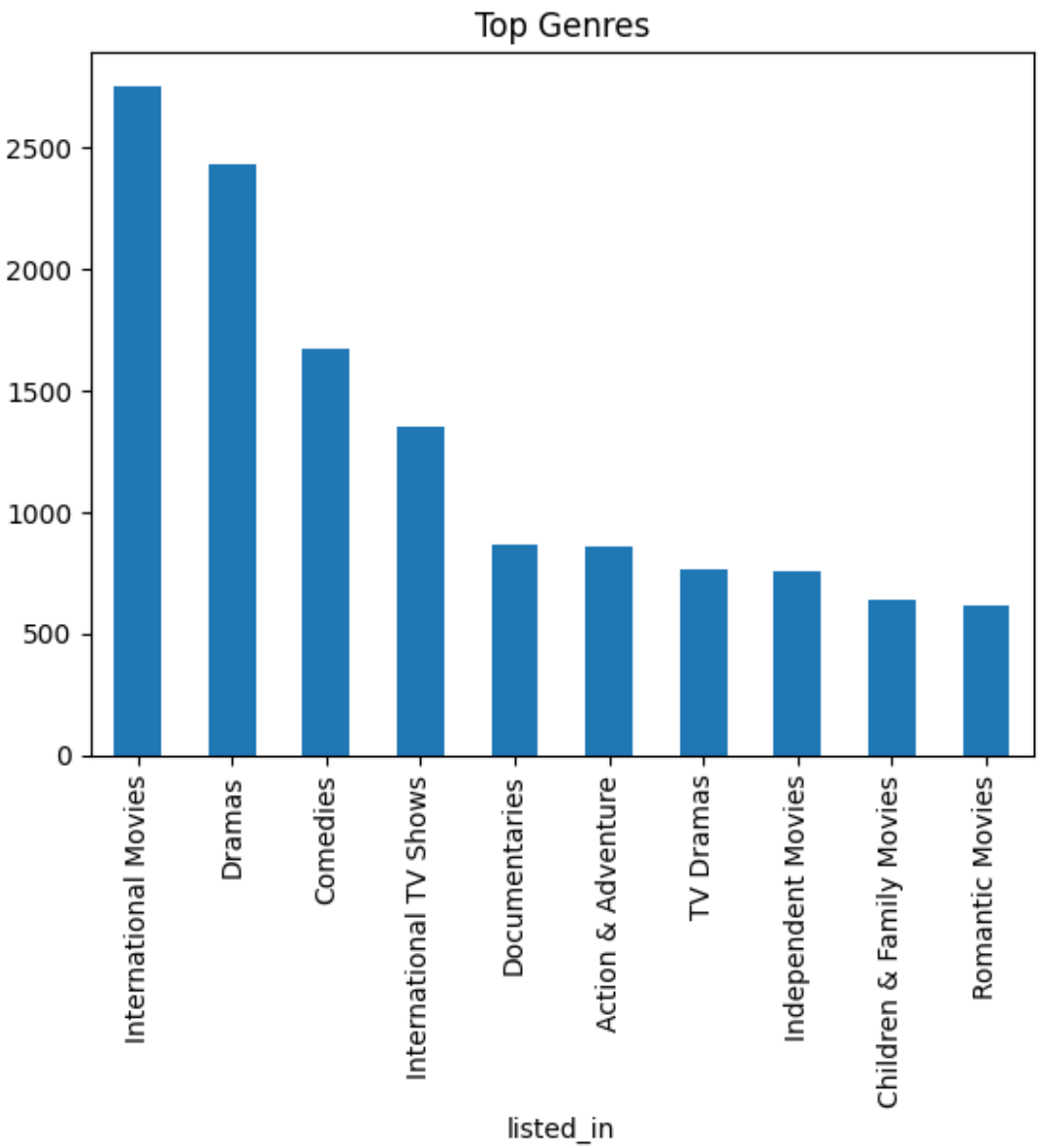
```
type
Movie      6131
```

```
TV Show    2676
Name: count, dtype: int64
```

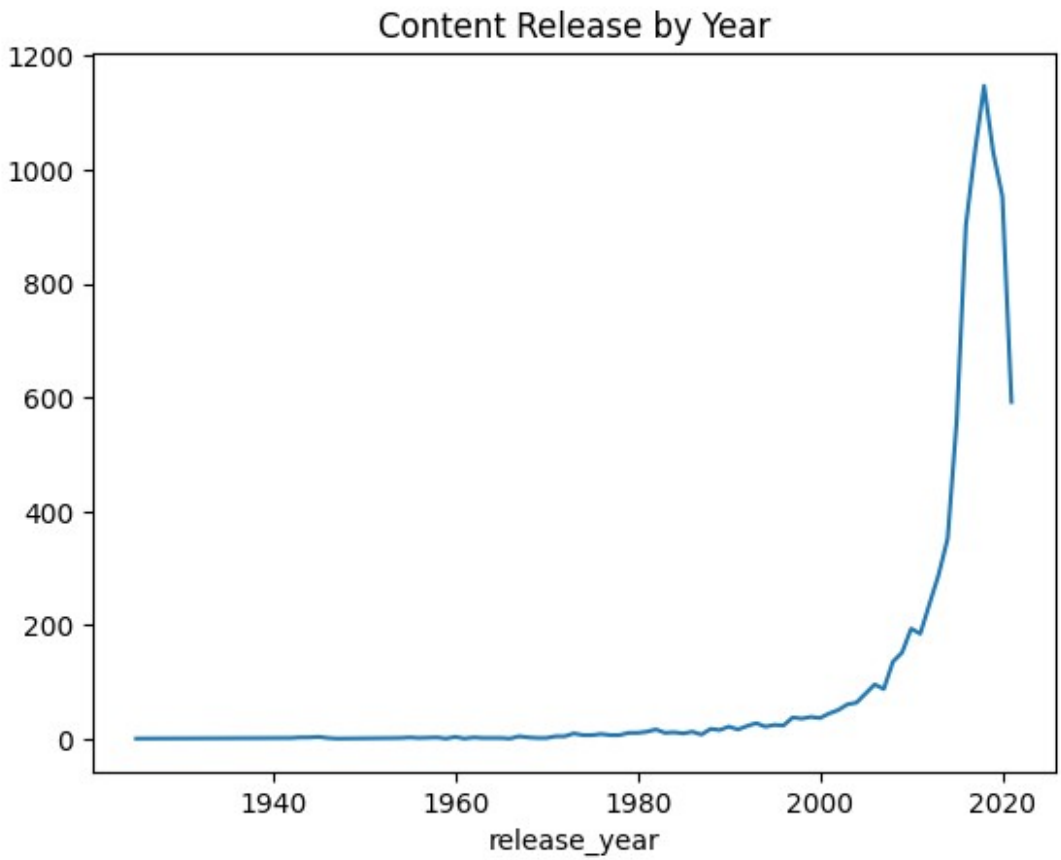



```
# Q2. Most popular genres globally
genres = df['listed_in'].str.split(', ').explode()
top_genres = genres.value_counts().head(10)
print(top_genres)

top_genres.plot(kind='bar')
plt.title("Top Genres")
plt.show()
```

```
listed_in
International Movies        2752
Dramas                      2427
Comedies                    1674
International TV Shows      1351
Documentaries                869
Action & Adventure           859
TV Dramas                    763
Independent Movies           756
Children & Family Movies     641
Romantic Movies              616
Name: count, dtype: int64
```

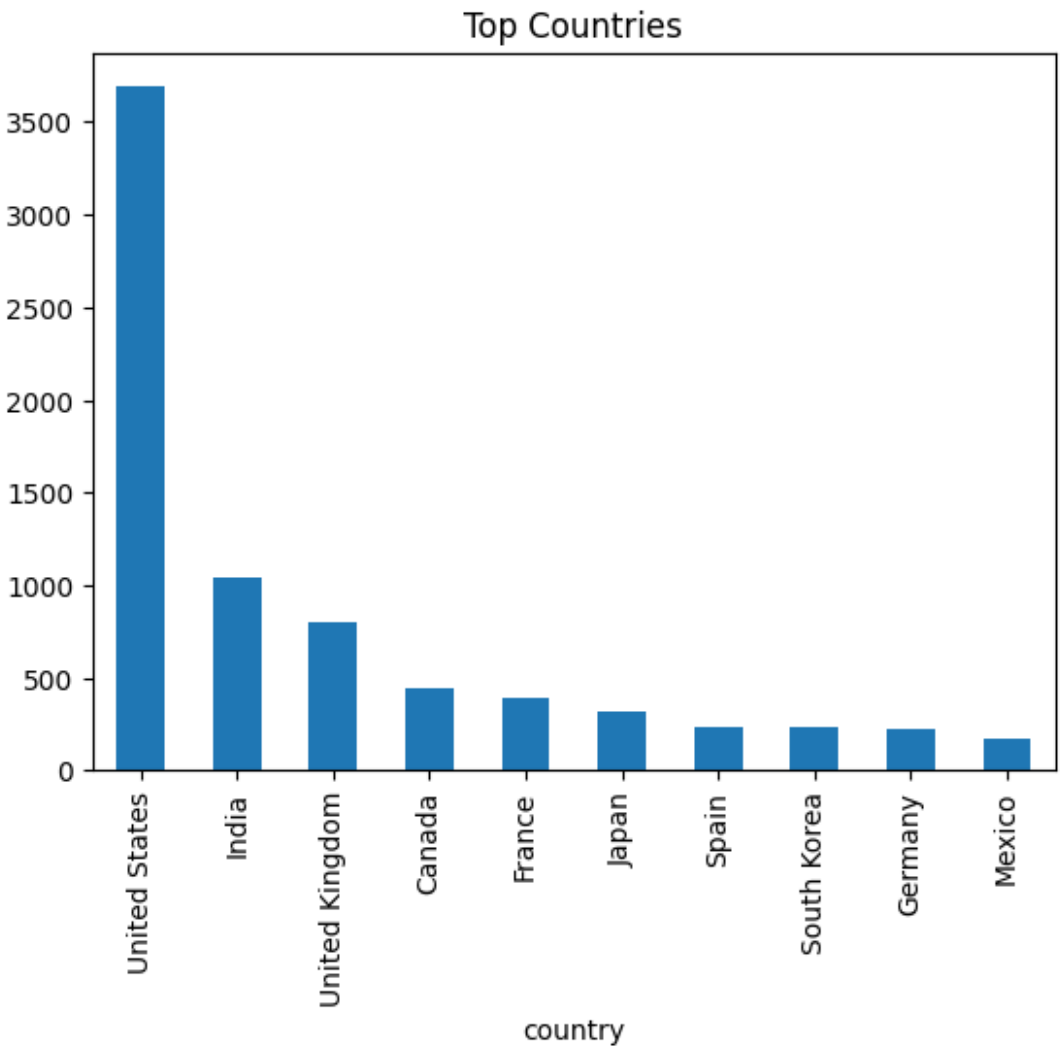



```
# Q3. Years with highest release content
release_year_counts = df['release_year'].value_counts().sort_index()
release_year_counts.plot()
plt.title("Content Release by Year")
plt.show()
```

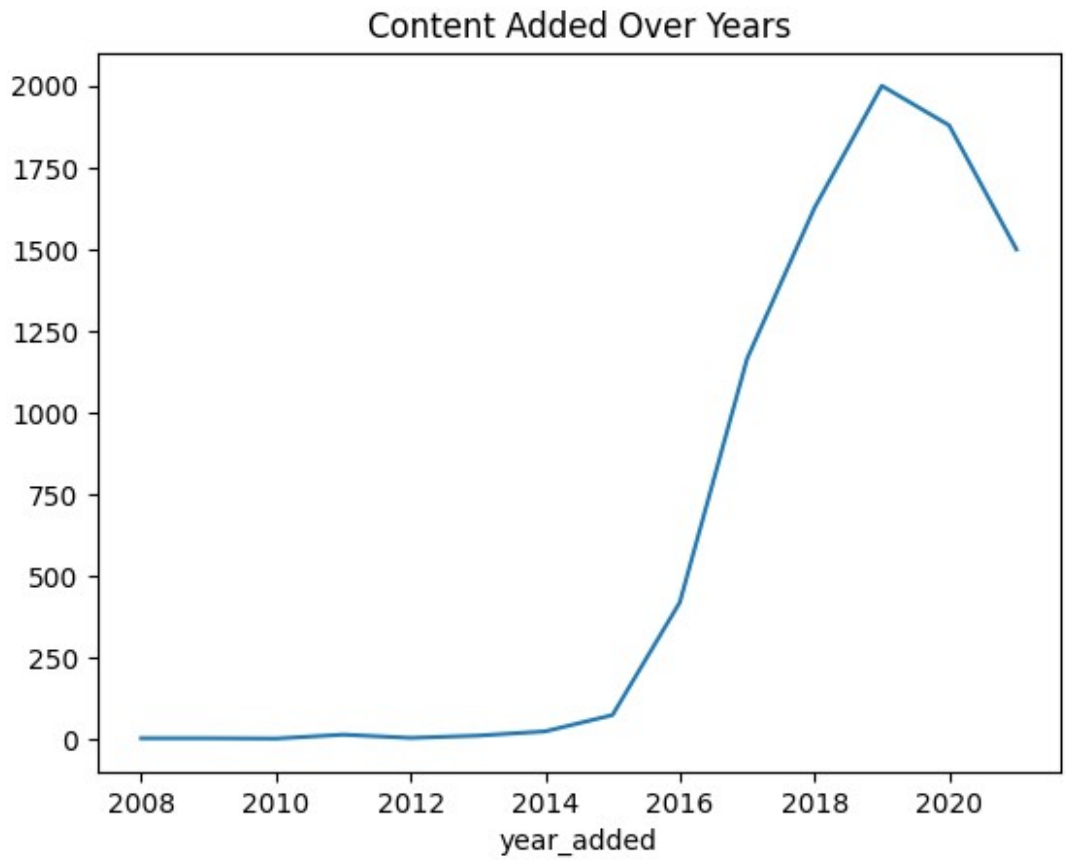



```python
# Q4. Countries producing most content
countries = df['country'].str.split(', ').explode()
top_countries = countries.value_counts().head(10)
print(top_countries)

top_countries.plot(kind='bar')
plt.title("Top Countries")
plt.show()
```

```
country
United States     3689
India             1046
United Kingdom     804
Canada             445
France             393
Japan              318
Spain              232
South Korea        231
Germany            226
Mexico             169
Name: count, dtype: int64
```




```
# Q5. Trend of adding content year by year
year_added_counts = df['year_added'].value_counts().sort_index()
year_added_counts.plot()
plt.title("Content Added Over Years")
plt.show()
```

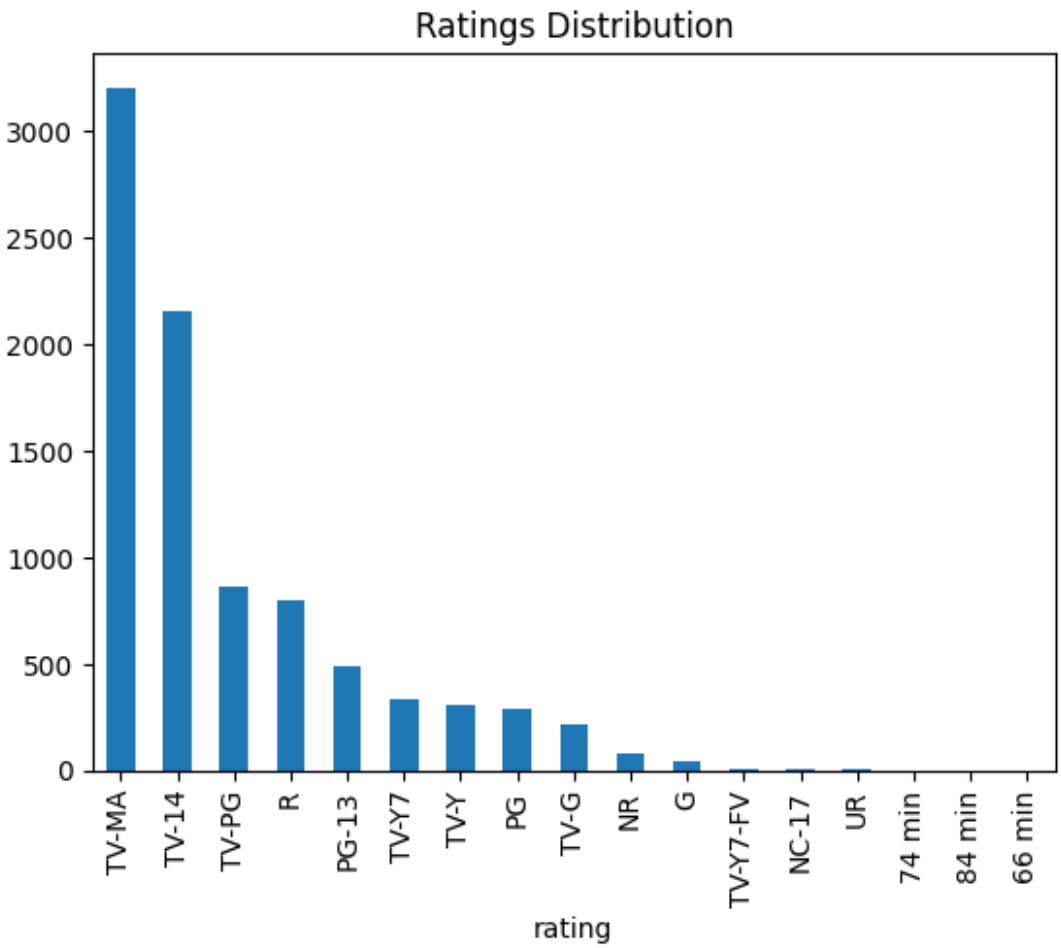


2. User Demographics & Targeting

```
# Q6. Most frequent ratings
ratings = df['rating'].value_counts()
print(ratings)

ratings.plot(kind='bar')
plt.title("Ratings Distribution")
plt.show()
```

```
rating
TV-MA       3207
TV-14       2160
TV-PG        863
R            799
PG-13        490
TV-Y7        334
TV-Y         307
PG           287
TV-G         220
NR            80
G             41
TV-Y7-FV       6
```

```
NC-17           3
UR              3
74 min          1
84 min          1
66 min          1
Name: count, dtype: int64
```




```python
# Q7. Countries producing more TV-MA content
tv_ma = df[df['rating'] == 'TV-MA']
countries_tv_ma = tv_ma['country'].str.split(',
').explode().value_counts().head(10)
print(countries_tv_ma)
```

```
country
United States     1100
India              266
United Kingdom     251
Spain              170
France             163
Canada             107
```

```
Mexico              102
Japan               101
South Korea          92
Germany              79
Name: count, dtype: int64
```

```
# Q8. Genres associated with Movies vs TV Shows
genre_type = df.copy()
genre_type['listed_in'] = genre_type['listed_in'].str.split(', ')
genre_type = genre_type.explode('listed_in')

pd.crosstab(genre_type['listed_in'], genre_type['type']).head(10)
```

```
type                       Movie  TV Show
listed_in
Action & Adventure           859        0
Anime Features                71        0
Anime Series                   0      176
British TV Shows               0      253
Children & Family Movies     641        0
Classic & Cult TV              0       28
Classic Movies               116        0
Comedies                    1674        0
Crime TV Shows                 0      470
Cult Movies                   71        0
```

```
# Q9. Genres in US vs other countries
df['is_US'] = df['country'].str.contains('United States', na=False)

us_genres = df[df['is_US']]['listed_in'].str.split(',
').explode().value_counts().head(10)
non_us_genres = df[~df['is_US']]['listed_in'].str.split(',
').explode().value_counts().head(10)

print("US Genres:\n", us_genres)
print("\nNon-US Genres:\n", non_us_genres)
```

```
US Genres:
 listed_in
Dramas                      835
Comedies                    680
Documentaries               512
Action & Adventure          404
Children & Family Movies    390
Independent Movies          390
Thrillers                   292
TV Comedies                 258
TV Dramas                   232
Romantic Movies             225
Name: count, dtype: int64
```

```
Non-US Genres:
 listed_in
International Movies      2586
Dramas                    1592
International TV Shows    1277
Comedies                   994
TV Dramas                  531
Action & Adventure         455
Romantic Movies            391
Independent Movies         366
Documentaries              357
Romantic TV Shows          326
Name: count, dtype: int64
```

```
# Q10. Popular genres in last 3 years
recent = df[df['release_year'] >= df['release_year'].max() - 3]
recent_genres = recent['listed_in'].str.split(',
').explode().value_counts().head(10)
print(recent_genres)
```

```
listed_in
International Movies        1002
Dramas                       834
International TV Shows       754
Comedies                     540
TV Dramas                    452
Documentaries                354
TV Comedies                  334
Crime TV Shows               305
Children & Family Movies     274
Independent Movies           269
Name: count, dtype: int64
```

3. Talent Analysis

```
# Q11. Top 10 directors
directors = df['director'].str.split(', ').explode()
top_directors = directors.value_counts().head(10)
print(top_directors)
```

```
director
Rajiv Chilaka          22
Jan Suter              21
Raúl Campos            19
Marcus Raboy           16
Suhas Kadav            16
Jay Karas              15
Cathy Garcia-Molina    13
Youssef Chahine        12
Martin Scorsese        12
```

```
Jay Chapman            12
Name: count, dtype: int64
```

```python
# Q12. Most frequent actors
actors = df['cast'].str.split(', ').explode()
top_actors = actors.value_counts().head(10)
print(top_actors)
```

```
cast
Anupam Kher         43
Shah Rukh Khan      35
Julie Tejwani       33
Takahiro Sakurai    32
Naseeruddin Shah    32
Rupa Bhimani        31
Om Puri             30
Akshay Kumar        30
Yuki Kaji           29
Amitabh Bachchan    28
Name: count, dtype: int64
```

```python
# Q13. Director-Genre pairs
df_pair = df.copy()
df_pair['listed_in'] = df_pair['listed_in'].str.split(', ')
df_pair = df_pair.explode('listed_in')

pairs = df_pair.groupby(['director',
'listed_in']).size().sort_values(ascending=False).head(10)
print(pairs)
```

```
director                listed_in
Rajiv Chilaka           Children & Family Movies    19
Raúl Campos, Jan Suter  Stand-Up Comedy             18
Suhas Kadav             Children & Family Movies    16
Marcus Raboy            Stand-Up Comedy             15
Jay Karas               Stand-Up Comedy             13
Cathy Garcia-Molina     International Movies        13
Jay Chapman             Stand-Up Comedy             12
Youssef Chahine         Dramas                      12
                        International Movies        10
Martin Scorsese         Dramas                       9
dtype: int64
```

```python
# Q14. Missing directors or cast
missing_director = df['director'].isna().sum()
missing_cast = df['cast'].isna().sum()

print("Missing Directors:", missing_director)
print("Missing Cast:", missing_cast)
```

```
Missing Directors: 2634
Missing Cast: 825
```

4. Duration & Engagement

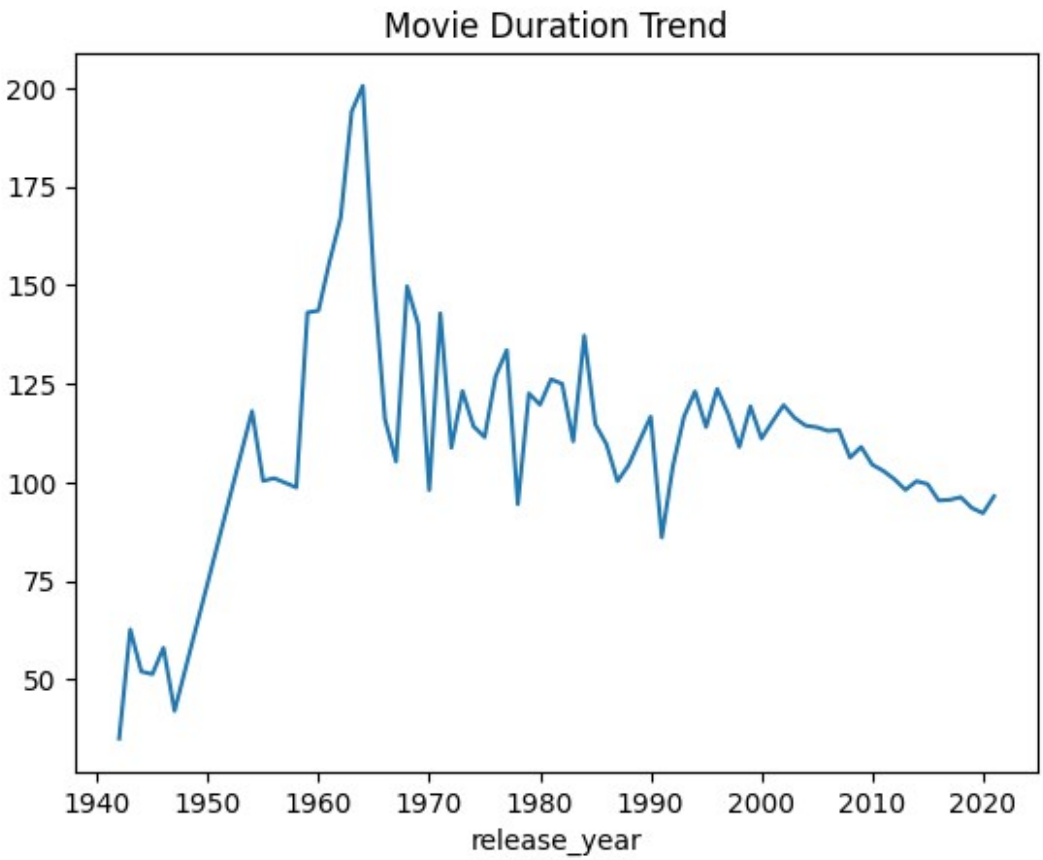
```
# Q15. Average movie duration
movies = df[df['type'] == 'Movie'].copy()

movies['duration'] = movies['duration'].str.replace(' min',
'').astype(float)
print("Average Duration:", movies['duration'].mean())
```

```
Average Duration: 99.57718668407311
```

```
# Q16. Most common number of seasons
tv = df[df['type'] == 'TV Show'].copy()
tv['duration'] = tv['duration'].str.replace(' Seasons',
'').str.replace(' Season', '')
print(tv['duration'].value_counts().head())
```

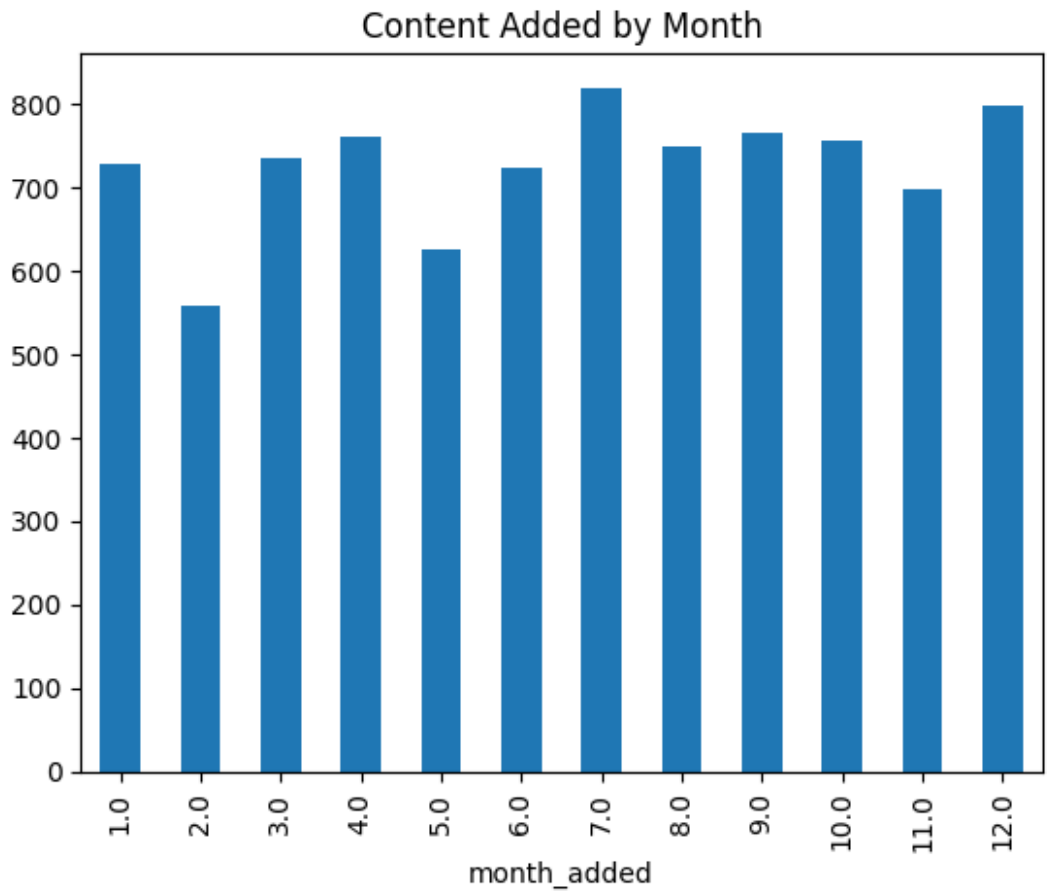
```
duration
1    1793
2     425
3     199
4      95
5      65
Name: count, dtype: int64
```

```
# Q17. Trend in movie duration over years
movies.groupby('release_year')['duration'].mean().plot()
plt.title("Movie Duration Trend")
plt.show()
```



5. Content Launch Strategy

```
# Q18. Month with most content added
month_counts = df['month_added'].value_counts().sort_index()
month_counts.plot(kind='bar')
plt.title("Content Added by Month")
plt.show()
```




```
# Q19. Genre distribution over years
genre_year = df.copy()
genre_year['listed_in'] = genre_year['listed_in'].str.split(', ')
genre_year = genre_year.explode('listed_in')

genre_year.groupby(['release_year',
'listed_in']).size().unstack().fillna(0).tail()
```

```
listed_in     Action & Adventure  Anime Features  Anime Series  \
release_year
2017                        89.0             6.0          10.0
2018                        81.0             8.0          24.0
2019                        44.0             6.0          18.0
2020                        46.0             3.0          21.0
2021                        37.0             6.0          23.0

listed_in     British TV Shows  Children & Family Movies  Classic &
Cult TV  \
release_year

2017                      34.0                      55.0
1.0
```

```
2018                        37.0                        69.0
2.0
2019                        26.0                        82.0
3.0
2020                        33.0                        83.0
3.0
2021                        17.0                        40.0
0.0

listed_in     Classic Movies  Comedies  Crime TV Shows  Cult
Movies  ...  \
release_year                                                     .
..
2017                      0.0     164.0            54.0
1.0  ...
2018                      1.0     178.0            79.0
1.0  ...
2019                      0.0     159.0            92.0
0.0  ...
2020                      0.0     133.0            87.0
0.0  ...
2021                      0.0      70.0            47.0
0.0  ...

listed_in     TV Action & Adventure  TV Comedies  TV Dramas  TV Horror
\
release_year

2017                            8.0         57.0       77.0        4.0

2018                           28.0         82.0      109.0       11.0

2019                           35.0         75.0      133.0       16.0

2020                           32.0        105.0      127.0       17.0

2021                           28.0         72.0       83.0       10.0


listed_in     TV Mysteries  TV Sci-Fi & Fantasy  TV Shows  TV
Thrillers  \
release_year

2017                   9.0                   4.0       2.0
3.0
2018                  15.0                   7.0       0.0
7.0
2019                  16.0                  14.0       2.0
16.0
2020                  27.0                  19.0       2.0
```

```
7.0
2021                  14.0                  14.0       2.0
9.0

listed_in     Teen TV Shows  Thrillers
release_year
2017                    5.0       68.0
2018                    8.0       83.0
2019                   14.0       71.0
2020                   11.0       45.0
2021                    8.0       33.0

[5 rows x 42 columns]
```

```python
# Q20. Countries producing most content per genre
country_genre = df.copy()
country_genre['country'] = country_genre['country'].str.split(', ')
country_genre['listed_in'] = country_genre['listed_in'].str.split(',
')

country_genre = country_genre.explode('country').explode('listed_in')

result = country_genre.groupby(['country',
'listed_in']).size().sort_values(ascending=False).head(10)
print(result)
```

```
country        listed_in
India          International Movies        864
United States  Dramas                      835
               Comedies                    680
India          Dramas                      662
United States  Documentaries               511
               Action & Adventure          404
               Independent Movies          390
               Children & Family Movies    390
India          Comedies                    323
United States  Thrillers                   292
dtype: int64
```

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Final Insights

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1. Movies dominate Netflix content over TV Shows.

2. Drama and International genres are most popular.

3. USA and India are top content producers.

4. TV-MA rating is most frequent → adult audience focus.

5. Content addition peaked after 2016 → rapid growth phase.

6. Most movies are around 90–120 minutes.

7. Netflix adds most content in last quarter months (Oct–Dec).

8. Recent years show rise in international content.